DFS Programs

1

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [], 'E': [], 'F': []
}

visited = []

def dfs(node):
    if node not in visited:
        visited.append(node)
        for child in graph[node]:
            dfs(child)

dfs('A')
print("DFS:", visited)
```

```
DFS: ['A', 'B', 'D', 'E', 'C', 'F']
```

2

```
graph = {
    '1': ['2', '3'],
    '2': ['4'],
    '3': ['5', '6'],
    '4': [], '5': [], '6': []
}

stack = ['1']
visited = []

while stack:
    node = stack.pop()
    if node not in visited:
        visited.append(node)
        stack.extend(reversed(graph[node]))

print("DFS:", visited)
```

```
DFS: ['1', '2', '4', '3', '5', '6']
```

3

```
graph = {
    'S': ['A', 'B'],
    'A': ['C', 'D'],
    'B': ['E'],
    'C': [], 'D': [], 'E': []
}

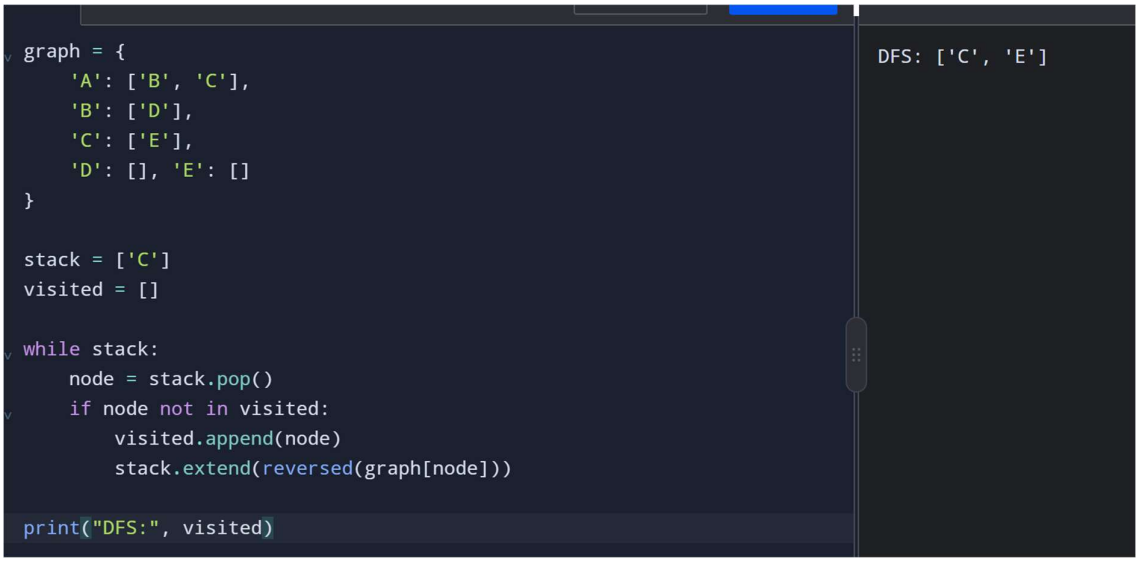
visited = []

def dfs(node):
    if node not in visited:
        visited.append(node)
        for child in graph[node]:
            dfs(child)

dfs('S')
print("DFS:", visited)
```

```
DFS: ['S', 'A', 'C', 'D', 'B', 'E']
```

4



```
graph = {
    'A': ['B', 'C'],
    'B': ['D'],
    'C': ['E'],
    'D': [], 'E': []
}

stack = ['C']
visited = []

while stack:
    node = stack.pop()
    if node not in visited:
        visited.append(node)
        stack.extend(reversed(graph[node]))

print("DFS:", visited)
```

```
DFS: ['C', 'E']
```